

PROXMOX VE STRETCHED CLUSTER

A Proxmox VE Stretched (Metro) Cluster is the ultimate solution for site-level disaster recovery and continuous uptime.

By extending your cluster and its shared storage across two physically separate data centers in a metropolitan area, you gain unparalleled resilience.

The core challenge is stretching the cluster's high availability and the VMs' shared storage across two separate physical locations while maintaining low latency.

CLUSTER QUORUM

A standard Proxmox VE cluster (based on Corosync) requires a majority of nodes to be online to maintain quorum and keep VMs running (to prevent a split-brain scenario), for this reason an odd number of votes is required.

For a two-site (Site A and Site B) stretched cluster:



Component	Quantity per site	Location	Purpose
Proxmox VE Nodes	5, 10, or 14... nodes	Sites A and B	Run Virtual Machines (VMs) and provide the cluster votes. Nodes must be split evenly between Site A and Site B.
QDevice	1 (Single, separate instance)	Third, Independent Location	Provides the single, crucial tie-breaker vote. Prevents split-brain by ensuring one site retains quorum if the inter-site link fails. Must have a stable connection to both sites.

Note: For a Ceph-based stretched cluster, utilizing a real Proxmox VE node in a third location is preferred over a QDevice. This design automates the management of the Ceph monitor agent, significantly simplifying the cluster's long-term operation. While it is technically possible to use a nested Proxmox VE installation (e.g., a Proxmox VE VM running on an entirely separate cluster) for the extra vote, this is highly situational and requires careful review. We recommend consulting with Proxmox enterprise support to ensure this approach meets the specific requirements and fault-tolerance goals of your individual use case.

SHARED STORAGE

Since a stretched cluster requires VMs to be able to instantly run on the surviving site without data loss, the data must be synchronously replicated between the two sites.

Synchronous replication is only required when achieving **Zero Downtime** (for disaster avoidance) and **Zero Data Loss** (in a disaster scenario) are strict requirements.

Otherwise, asynchronous replication offers a simpler and significantly less resource-intensive alternative.



The most common ways to achieve this with Proxmox VE are:

Method	Description	Replication Type
Ceph Storage	Proxmox VE has native Ceph integration. You can configure a single Ceph cluster that spans both data centers, using Ceph's Stretch Mode and custom CRUSH rules to ensure data has replicas at both sites.	Synchronous
External SAN	If you have existing, dedicated storage they often can provide a hardware-based stretched-storage solution that handles the synchronous replication. Proxmox VE would then connect to this logical volume as shared storage.	Synchronous
ZFS	Proxmox VE integrates ZFS replication, which can also be leveraged for a two-site deployment. While it doesn't offer the same level of data protection as other solutions, its advantages lie in its low resource usage and straightforward management.	Asynchronous

NETWORK & LATENCY REQUIREMENTS

The cluster's stability is completely dependent on a low-latency network link, as Proxmox VE's core service, Corosync (for quorum) is highly sensitive to delay.

For a stretched cluster to be truly reliable, the network traffic must be physically separated to prevent congestion on one link from disrupting cluster heartbeat.

Service	Requirement	Impact of High Latency
Corosync (Quorum)	< 5 milliseconds (ms) RTT (Round-Trip Time) strongly recommended. < 10 milliseconds (ms) RTT is the absolute maximum limit for any cluster operation.	Cluster becomes unstable; nodes may be incorrectly flagged as failed and fence (reboot).
Dedicated NICs	2+ dedicated NICs per node (10+ Gbit/s recommended)	Isolation is key: One set of NICs for low-latency Corosync/Quorum traffic.

Some rough estimates for planning your deployments:

Round-Trip Time (RTT)	Estimated Max Distance (km)	Context
< 1 ms	~10 km	Data centers that are very close, possibly within the same campus or metropolitan area.
5 ms to 10 ms	~100 km to 200 km	Isolation is key: One set of NICs for low-latency Corosync/Quorum traffic.

Component	Minimum Specification	Purpose
Corosync	2 dedicated NICs per Node¹	Low latency for the most reliable connection possible
Storage (Ceph)	at least dedicated 10Gbps²	Stable high bandwidth connection between both sites for fast rebalancing in the case of an outage.
Storage (Other)	-	Please consult your storage vendor.

GET STARTED

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¹https://pve.proxmox.com/pve-docs/pve-admin-guide.html#pvecm_cluster_network_requirements
²https://pve.proxmox.com/pve-docs/pve-admin-guide.html#_recommendations_for_a_healthy_ceph_cluster